

Predictive Analytics Using Historical Data - Python Code

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

data={'Year':[2018,2019,2020,2021,2022,2023], 'Sales':[120,135,150,165,180,200]}
df=pd.DataFrame(data)
X=df[['Year']]
y=df['Sales']
model=LinearRegression()
model.fit(X,y)
future_years=np.array([[2024],[2025],[2026]])
predicted_sales=model.predict(future_years)
print(predicted_sales)
plt.figure(figsize=(8,6))
plt.scatter(X,y, label='Historical Data')
plt.plot(X,model.predict(X), label='Regression Line')
plt.scatter(future_years,predicted_sales,marker='*',s=200, label='Predicted Sales')
plt.title('Predictive Analytics Using Historical Data')
plt.xlabel('Year')
plt.ylabel('Sales')
plt.legend()
plt.grid(True)
plt.show()
```